We see that

$$I(u_{n+1}, v_{n+1}) = u_{n+1} - v_{n+1} = u_n - v_n = I(u_n, v_n),$$

which means that Euler's method conserves this first integral. We say that this conservation property is *exact*, even though, in practice, there could be fluctuations in the result depending on how the computations are carried out on a computer (normally, if the relevant algebraic equations are solved numerically, there will be small residual errors which are similar in magnitude to the computer rounding error.

Generalizing this slightly, we could imagine a system of ODEs of the form

$$\dot{\boldsymbol{z}} = \boldsymbol{f}(\boldsymbol{z}),$$

such that, for some vector $\boldsymbol{b}$,

$$\boldsymbol{b} \cdot \boldsymbol{f}(\boldsymbol{z}) \equiv 0,$$

then $I(\boldsymbol{z}) = \boldsymbol{b} \cdot \boldsymbol{z}$ is a first integral. It is straightforward to see that Euler's method conserves such a *linear* first integral exactly, as is true of many other popular numerical methods. Thus momentum conservation is rarely violated by (sensible) numerical methods.

The interest is usually in conserving some more complicated first integrals, for example the energy of the system. None of the schemes we have considered so far (Euler's method, Störmer-Verlet, etc.) conserves this quantity exactly, even in the case of a harmonic oscillator. In the case of Euler's method applied to the harmonic oscillator, the error in energy grows with time and without bound. In the case of Störmer-Verlet, the energy fluctuates but remains bounded for all time and at its worst is of size proportional to h^2, a numerical observation that is supported by the existence of the modified Hamiltonian.

It is tempting to consider energy-conserving methods by which the energy is corrected at each step back to the initial value. This is easy to do in general, and there are in fact many choices of the method used. Let us consider one particular approach. We start by taking a timestep with any arbitrary (nonconserving) numerical method, say to an intermediate phase point $\bar{\boldsymbol{Q}}, \bar{\boldsymbol{P}}$, then we "fix it up" by scaling the momentum by an adjustment factor, defining

$$\boldsymbol{Q} = \bar{\boldsymbol{Q}}, \qquad \boldsymbol{P} = \gamma \bar{\boldsymbol{P}},$$

selecting γ so that the energy in the result is a prescribed value E. The energy fluctuation in our pre-correction scheme would typically be small in a single timestep, so the correction γ should also be close to one. This method is easy to implement: the equation that must be solved is

$$\frac{\gamma^2 \bar{\boldsymbol{P}}^T \boldsymbol{M}^{-1} \bar{\boldsymbol{P}}}{2} + U(\bar{\boldsymbol{Q}}) = \mathrm{E}, \tag{3.12}$$